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Supplementary Materials for

Structure of the cytoplasmic ring of the *Xenopus laevis* nuclear pore complex

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The PDF file includes:

Figs. S1 to S18
Tables S1 to S5

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

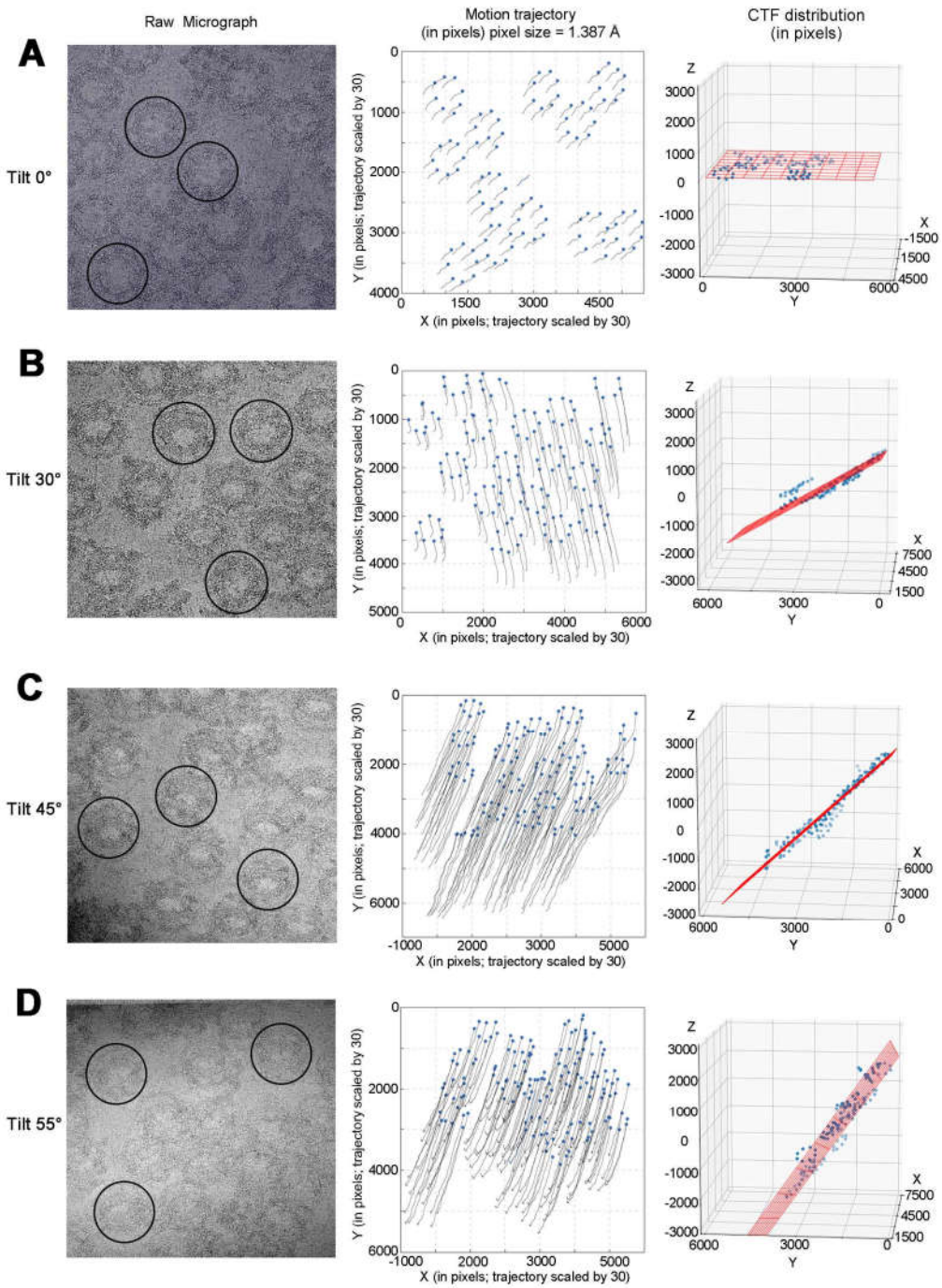


Fig. S1.

Analysis of the quality of the cryo-EM data for the NPC from *Xenopus laevis* (*X. laevis*) oocytes. (**A-D**) Representative micrographs from the datasets for tilt-0° (**A**), tilt-30° (**B**), tilt-45° (**C**), and tilt-55° (**D**) and their quality assessment. A summed micrograph with dose weighting applied, the motion trajectory, and the CTF distribution of the CR subunit core region are shown on the left, middle, and right, respectively, in each panel. Compared to the tilt-0° dataset, micrographs of the tilt-30°, tilt-45° and tilt-55° datasets display increasingly large degrees of motion and wide distribution of per-particle defocus values.

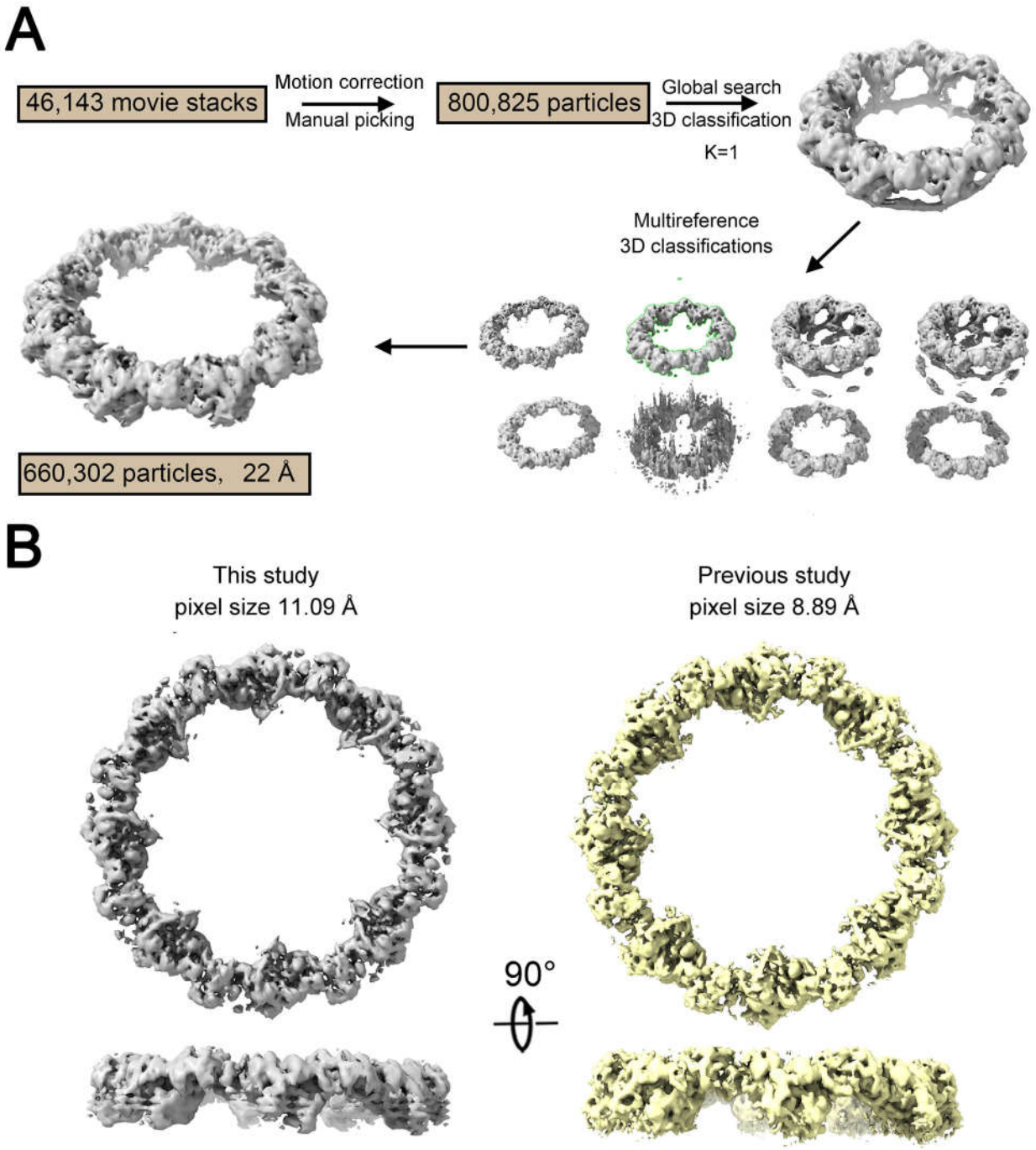


Fig. S2.

Preliminary processing of the cryo-EM data for the reconstruction of the intact CR from *X. laevis* NPC. (**A**) Flowchart of data processing for the reconstruction of the CR. For details, please refer to the section "Initial model of the cytoplasmic ring" in Material and Methods. (**B**) Comparison of the 3D EM reconstructions of the *X. laevis* CR with (present study) and without (our previous report (*14*)) crosslinking treatment of the nuclear envelope. Two representative views are shown. The overall structural features are nearly identical to each other.

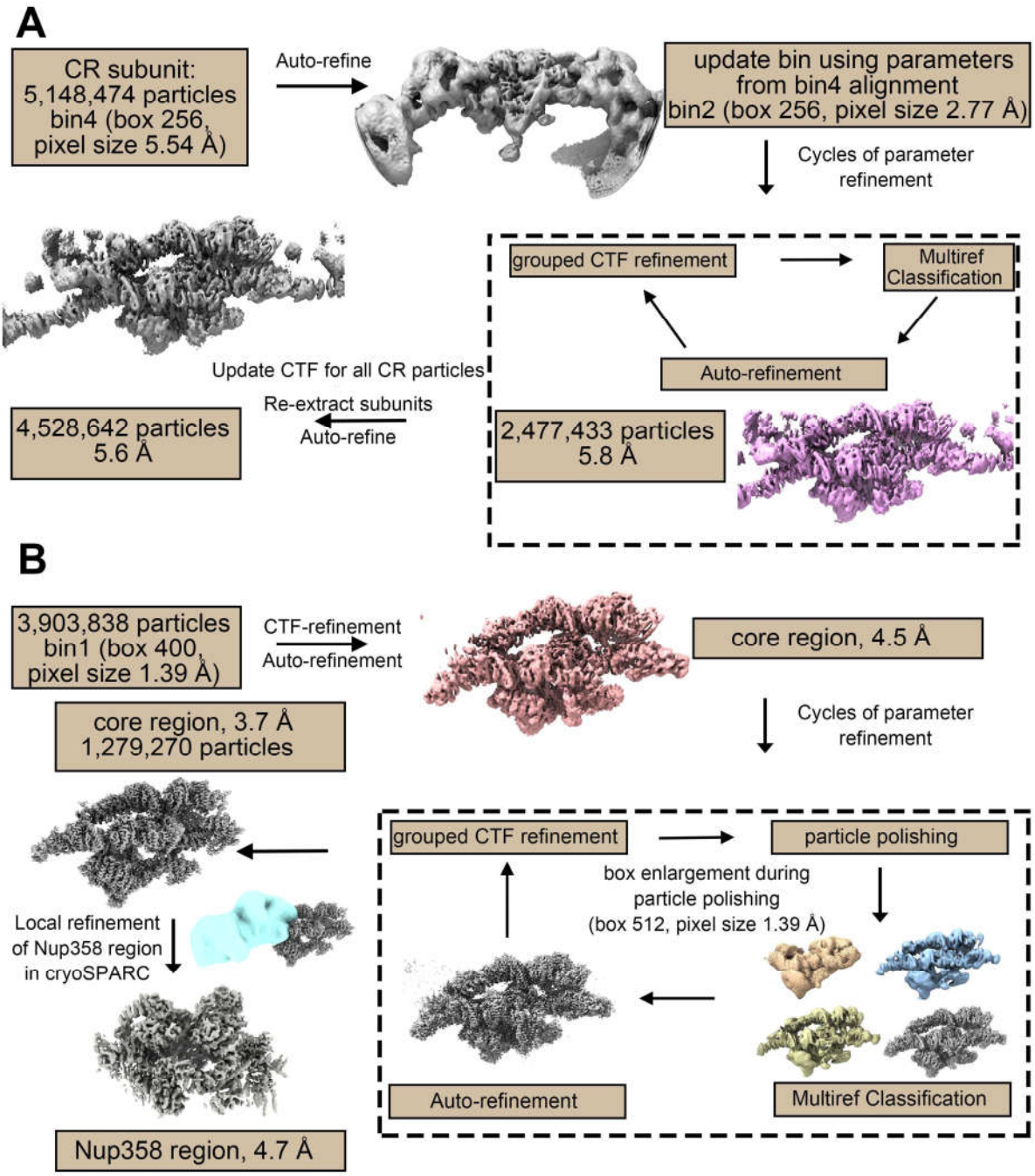


Fig. S3.

Cryo-EM data processing for the reconstruction of the CR subunit from *X. laevis* NPC. (**A**) Flowchart of bin2-level data processing for the reconstruction of the core region of the CR subunit. The core region was defined based on its central position and relative structural stability in the CR subunit. The remaining visible and continuous chunk was defined as the Nup358 region. Please refer to Fig. 1B for the approximate demarcation of the two regions. This practice resulted in a reconstruction of the core region at an average resolution of 5.6 Å. (**B**) Flowchart of bin1-level data processing for the core region and the Nup358 region. For details, please refer to the section "Data processing and reconstruction of the CR subunit" in Material and Methods.

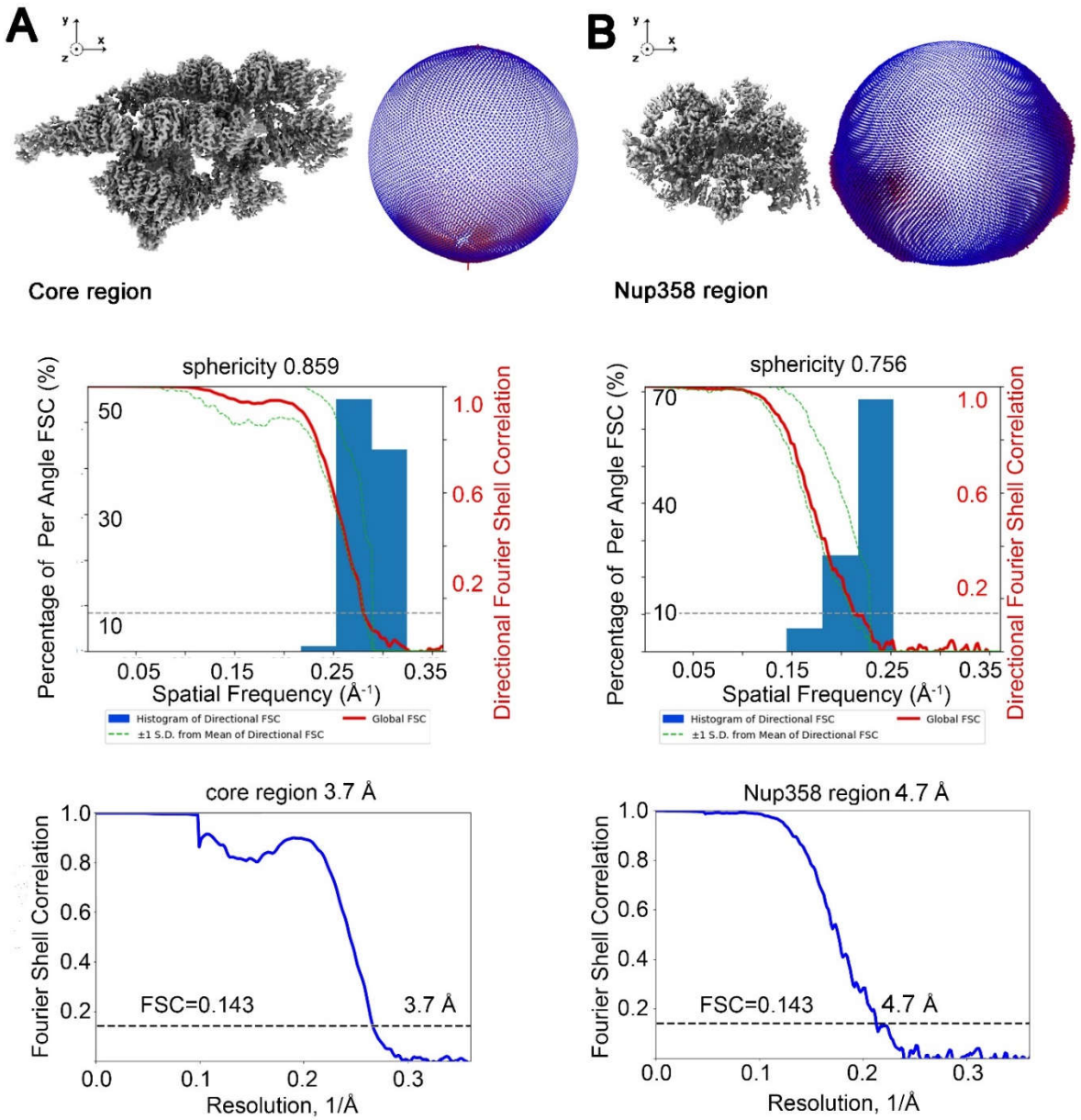


Fig. S4.

Cryo-EM reconstruction of the core region (A) and the Nup358 region (B). Angular distribution of the particles used for the final reconstructions, the directional FSC, and the final FSC curves are shown in the top, middle and bottom panels, respectively. All directional FSC curves were prepared using https://3dfsc.salk.edu (*40*).

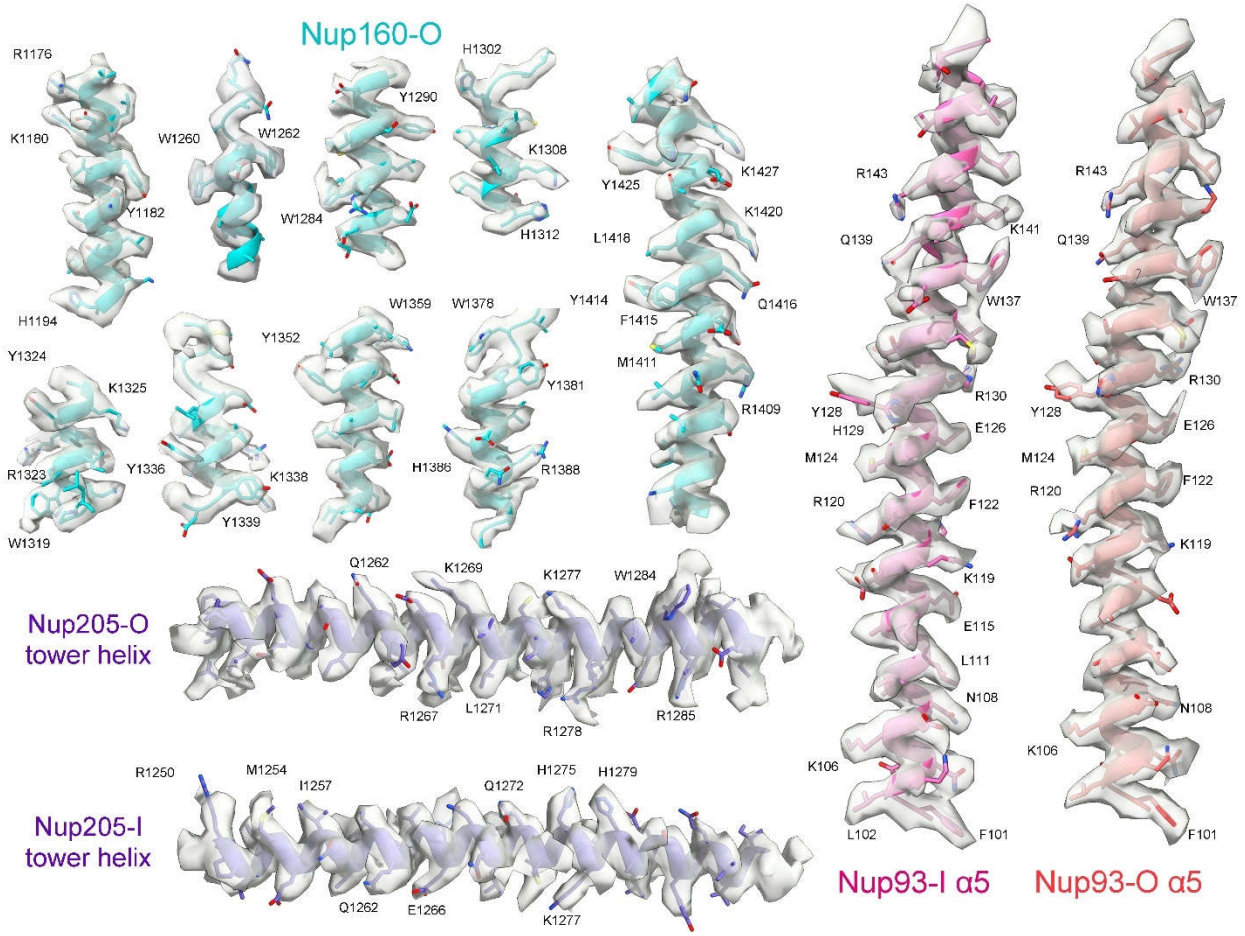


Fig. S5.

Representative EM maps for the CR subunit. EM maps for representative segments from the C-terminal fragment (CTF) of Nup160-O, two Nup205 molecules and two Nup93 molecules. Components in the outer and inner Y complexes are annotated with "O" and "I", respectively. All EM maps in this figure were prepared using the reconstruction of the core region with a contour level between 5 σ and 10 σ in ChimeraX. To prepare the figures for local densities, the "volume zone" command in ChimeraX was used to selectively display EM densities in the vicinity of the selected atoms and the "hide dust" command was used to remove noises in the EM density.

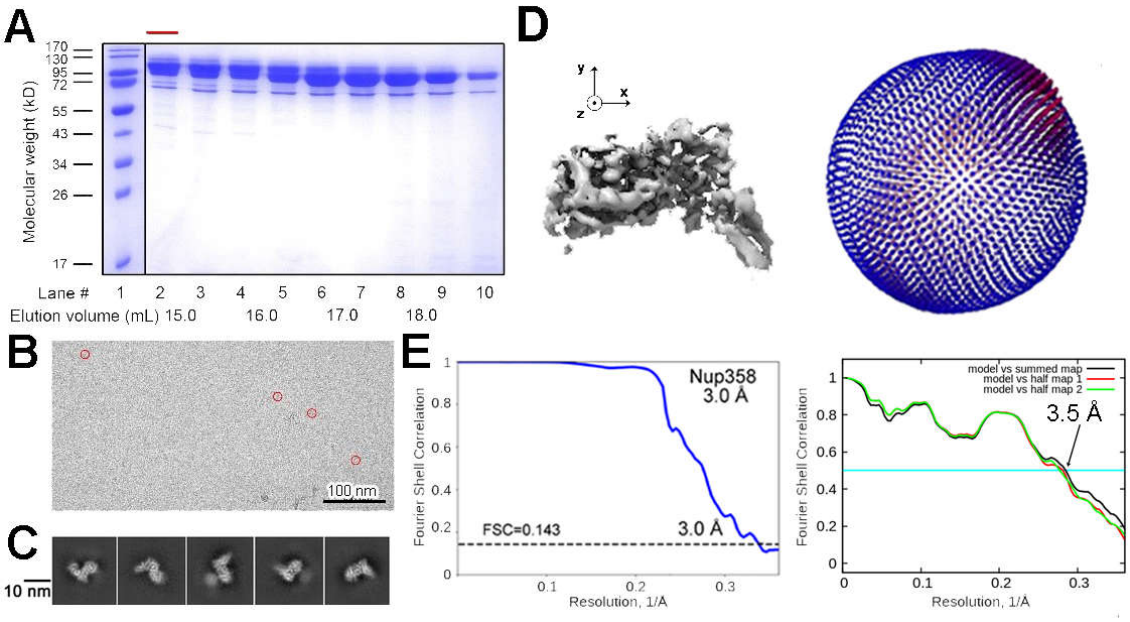


Fig. S6.

Cryo-EM analysis of the N-terminal fragment (NTF) of Nup358. (**A**) Purification of Nup358-NTF. Following gel filtration, the peak fractions were resolved on SDS-PAGE and visualized by Coomassie blue staining. The expressed fragment contains residues 1-1171 of Nup358. Protein degradation during purification resulted in fragments of smaller molecular weight. The fraction corresponding to lane 2 (indicated by red bar) was used for cryo-sample preparation. (**B**) A representative EM micrograph of Nup358-NTF. (**C**) Representative 2D class averages of Nup358-NTF. Scale bar, 10 nm. (**D**) Angular distribution of the final reconstruction. (**E**) The FSC curves. Shown on the left is the final FSC curve for Nup358-NTF, and on the right are the FSC curves of the final reconstruction against the refined atomic coordinates. The coordinates were first refined against the map from the whole data set and the map-vs-model FSC was calculated against the first half map (red), second half map (green) and final map (black). Small differences among the three FSC curves suggest that there is no apparent overfitting in the final model.

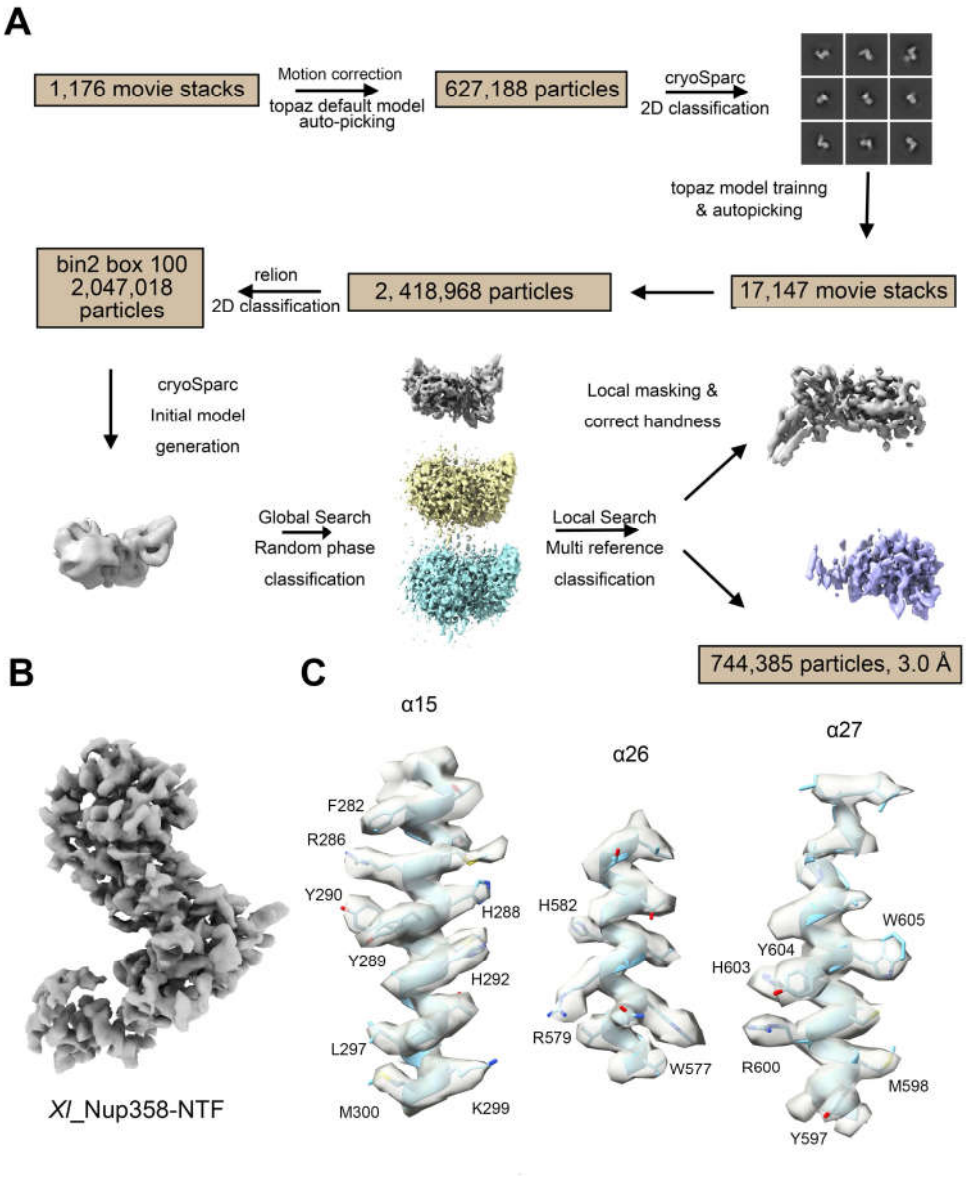


Fig. S7.

Cryo-EM data processing and representative EM maps for Nup358-NTF. (**A**) Workflow for cryo-EM analysis of Nup358-NTF from *X. laevis*. For details, please refer to the section "Image processing for Nup358-NTF" in Materials and Methods. (**B**) Overall EM map of Nup358-NTF. The density map was prepared in ChimeraX with a contour level of 10 σ. (**C**) Representative segments of the EM map of Nup358-NTF. Bulky side chains that facilitated sequence assignment are labeled. To prepare the figure, the "volume zone" command in ChimeraX was used to selectively display voxels that are within 2.5-Å range of the displayed atoms. The "hide dust" command was applied with dust size at 5. The contour level for the left and middle panel is 10 σ. The contour level for the right panel is set to 14 σ.

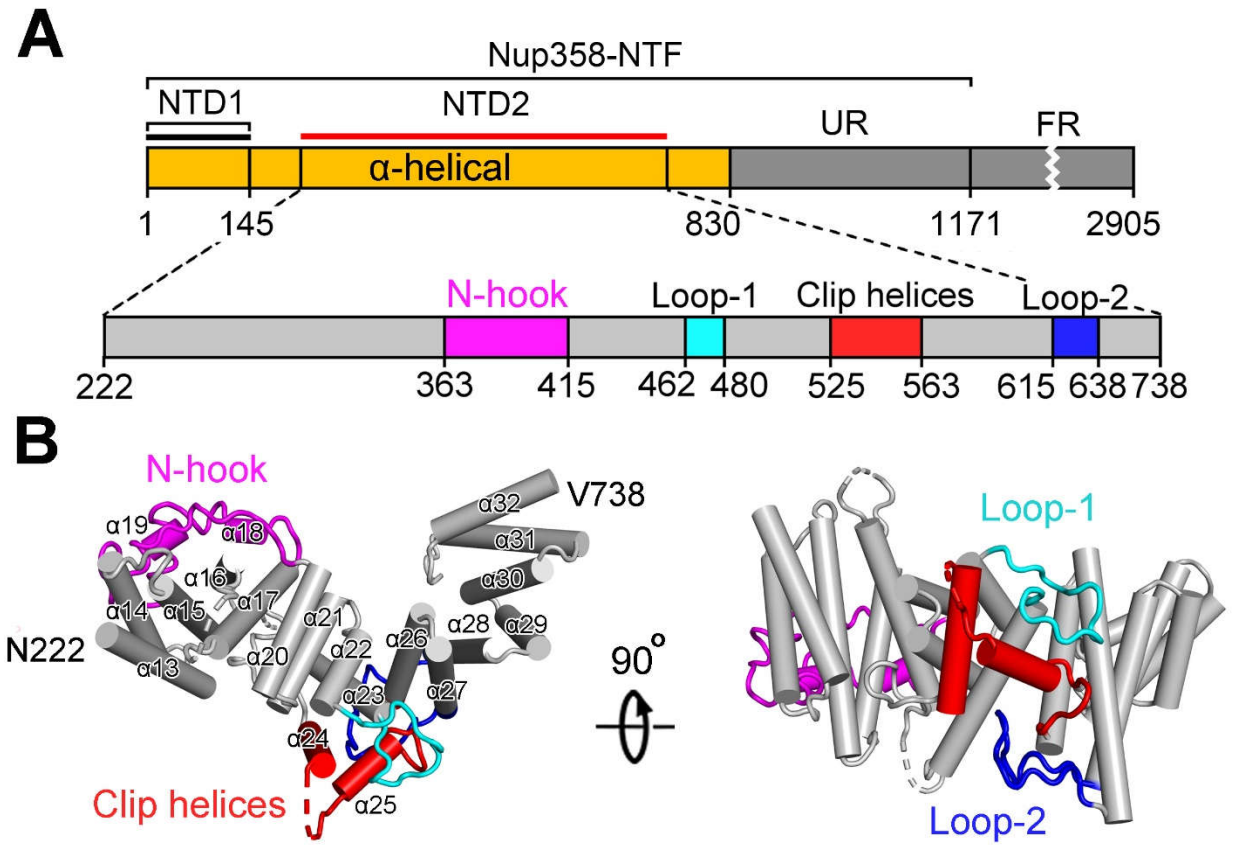


Fig. S8.

Structure of Nup358-NTD2. (**A**) The topological structure of Nup358-NTF. Nup358-NTF contains NTD1 and NTD2. NTD2 (residues 222-738) is well-defined in the EM map of Nup358-NTF. Four distinct structural motifs, N-hook, Loop-1, Clip helices, and Loop-2, are highlighted. (**B**) Overall structure of Nup358-NTD2. Two perpendicular views are shown.

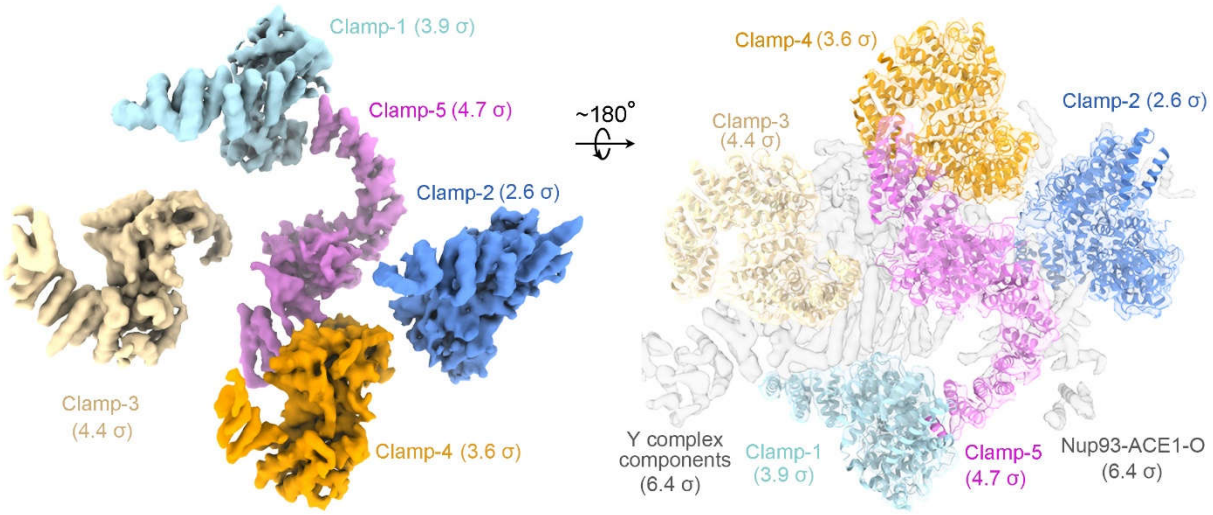
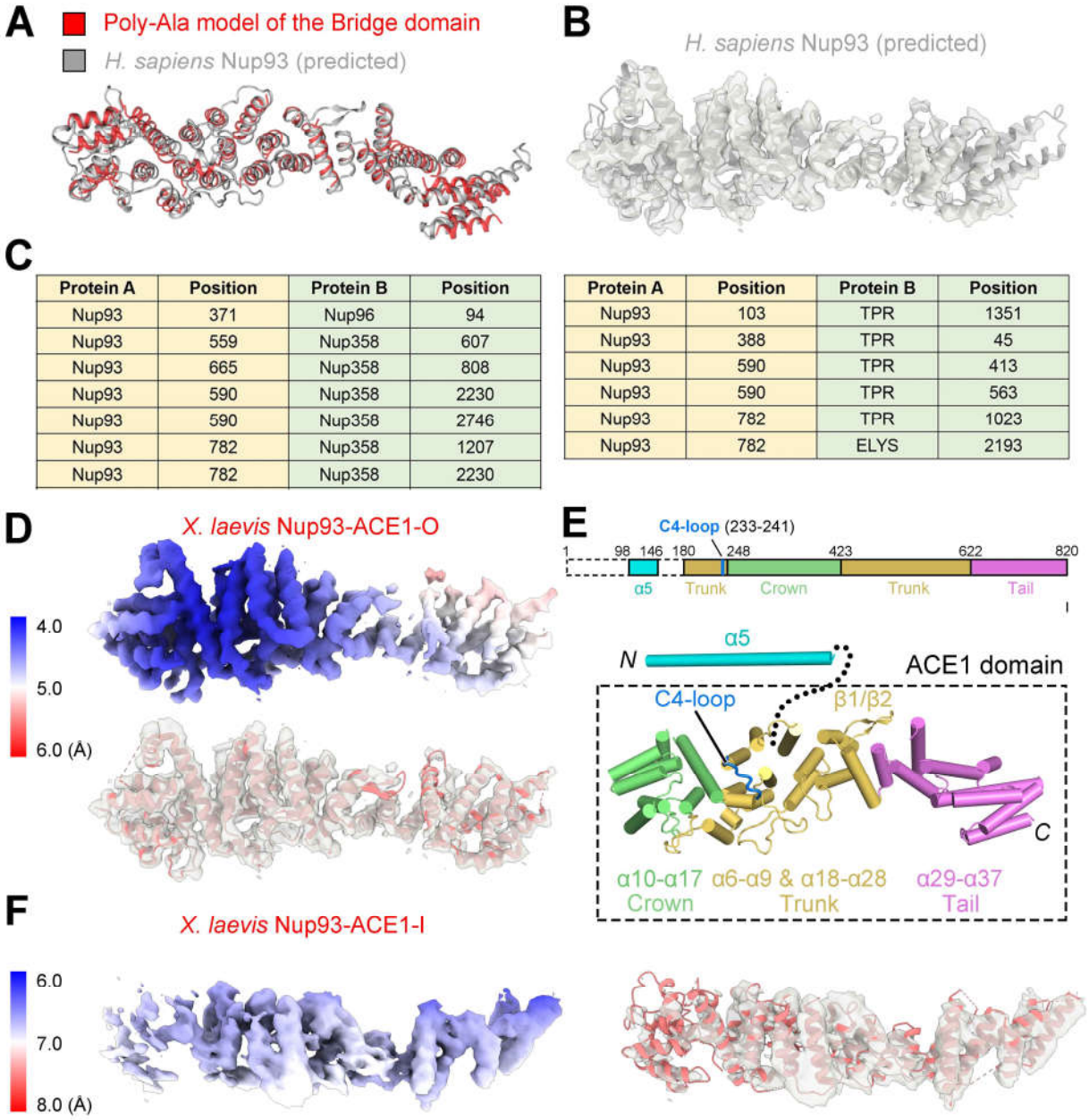


Fig. S9.

EM maps for five Nup358 clamps. Five Nup358 molecules are revealed in the EM map. *Left*: Color-coded maps for the five clamps. *Right*: Structures of Nup358 molecules docked into these maps, presented as semi-transparent surface. The two views are related by a rotation of approximately 180 degrees. To generate the figure, densities corresponding to Nup358 molecules were segmented from locally refined 4.7-Å EM map of the Nup358 region, using the map segmentation tool in Chimera. For visual clarity, different sigma levels, indicated in the brackets, are used for each clamp map.



Protein A	Position	Protein B	Position
Nup93	371	Nup96	94
Nup93	559	Nup358	607
Nup93	665	Nup358	808
Nup93	590	Nup358	2230
Nup93	590	Nup358	2746
Nup93	782	Nup358	1207
Nup93	782	Nup358	2230

Protein A	Position	Protein B	Position
Nup93	103	TPR	1351
Nup93	388	TPR	45
Nup93	590	TPR	413
Nup93	590	TPR	563
Nup93	782	TPR	1023
Nup93	782	ELYS	2193

Fig. S10.

Identification of the Bridge domain as Nup93. (**A**) Superimposition of the poly-Ala model of the previously defined Bridge domain (*14*) and the predicted structure of human Nup93 from the database released by DeepMind (*54*). (**B**) Docking of the predicted structure of human Nup93 into the EM map of the Bridge domain. The EM map is derived from the 4.7-Å reconstruction for the Nup358 region and contoured of at 4 σ. (**C**) Mass spectrometric analysis of the crosslinked *X. laevis* nuclear envelope supports Nup93 to be the Bridge domain. Shown here are results of mass spectrometric analysis following chemical crosslinking of *X. laevis* NPC using glutaraldehyde. (**D**) Docking of the Alphafold2-predicted structure of *X. laevis* Nup93 (*15*) into the EM map of the Bridge domain. (**E**) Domain structure of *X. laevis* Nup93. The ACE1 domain is indicated by dashed box in the lower panel. (**F**) Docking of the structural model of Nup93-ACE1-O into the EM map for the second copy of Nup93, designated Nup93-ACE1-I. For panels

D & F, the densities were prepared with the locally refined maps for Nup93-ACE1-O and -I contoured at 4 σ and 3 σ, respectively. All density figures were prepared in ChimeraX, where the EM maps were zoned to 3 Å around the structural models and the "hide dust" command was applied with dust size at 5.

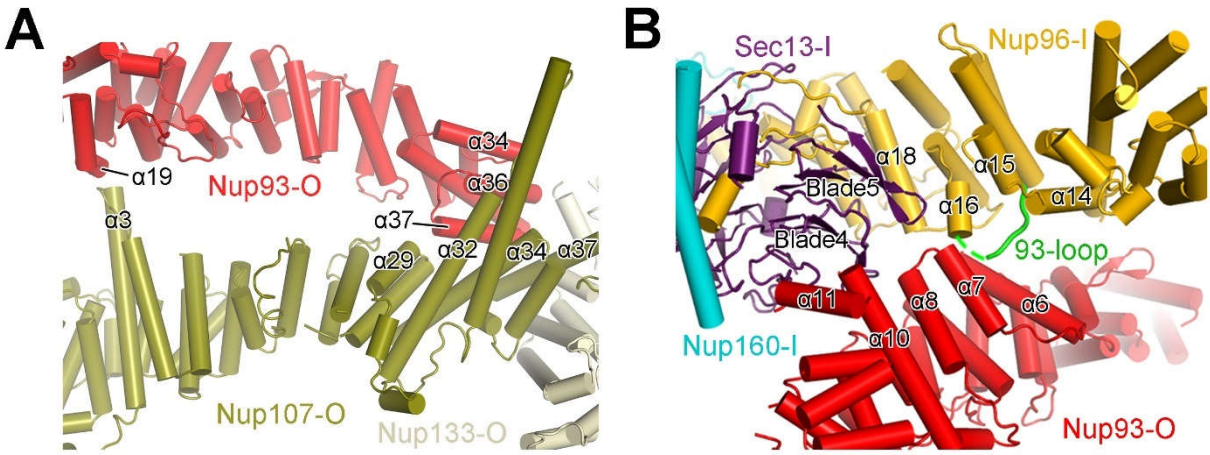


Fig. S11.

Association of Nup93-ACE1-O with other ACE1 proteins in the Y complex. (**A**) The interface of Nup93-ACE1-O with Nup107-O. On one end, α19 from Nup93-ACE1-O interacts with α3 from Nup107-O. On the other end, helices α34/36/37 from the Tail of Nup93-ACE1-O associate with helices α29/32/34/37 from the Tail of Nup107-O. (**B**) The interface between Nup93-O and Nup96-I/Sec13-I. A conserved loop between α15 and α16 of Nup96-I (referred to as 93-loop) interacts with the helical ridge of Nup93-O through binding to α6/7 of Nup93-ACE1-O. The Nup93-Nup96 association appears to be strengthened by Sec13-I, which co-folds with Nup96-I and uses its blades 4/5 to contact helices α11/α10 from the Crown of Nup93-O.

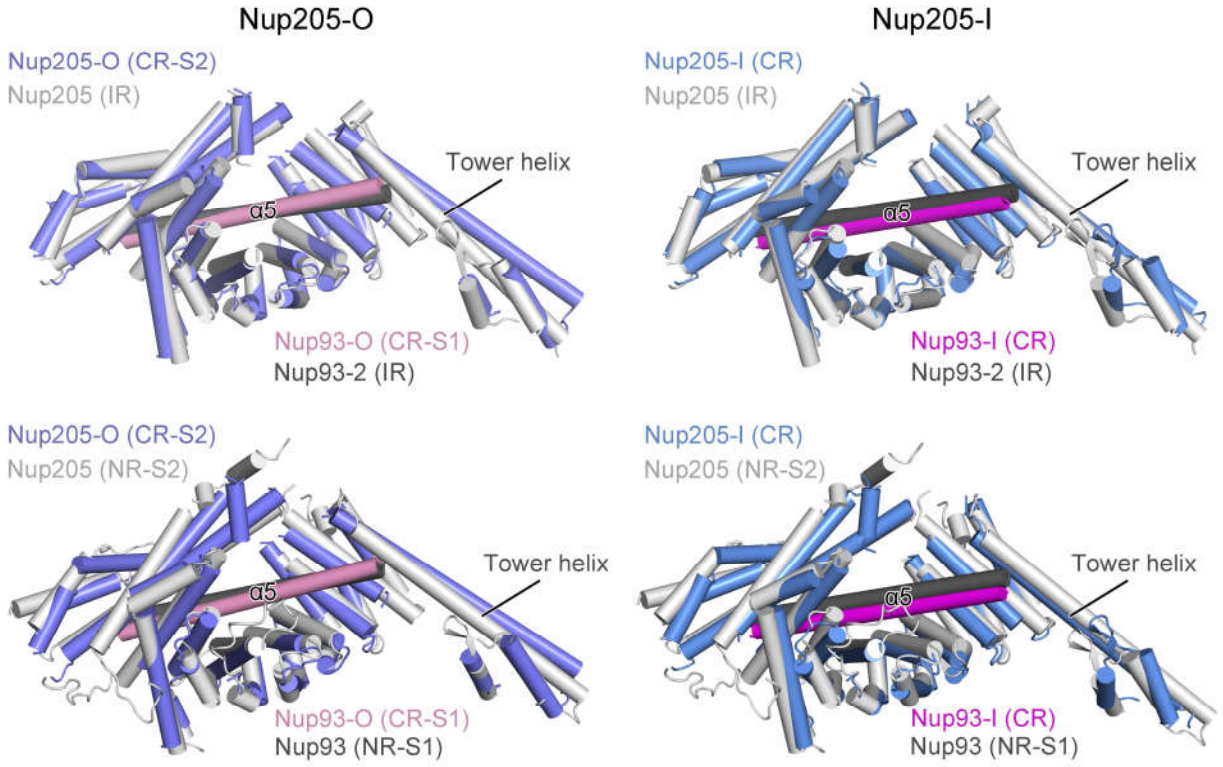


Fig. S12.

A conserved interaction between Nup93-α5 and Nup205-CTD in different NPC rings. Shown here are comparisons of the *X. laevis* Nup205-CTD (plus Tower helix)/Nup93-α5 complexes from the CR, IR and NR subunits. Nup205-O-CTD (S2)/Nup93-O-α5 (S1) from the CR can be superimposed with the counterparts from the IR and NR with RMSD of 2.3 Å and 3.4 Å over 571 and 565 aligned Cα atoms, respectively. Nup205-I-CTD/Nup93-I-α5 from the CR can be superimposed with the counterparts from the IR and NR with RMSD of 2.0 Å and 3.4 Å over 572 and 565 aligned Cα atoms, respectively. S1 and S2 refer to the neighboring subunits in the indicated ring.

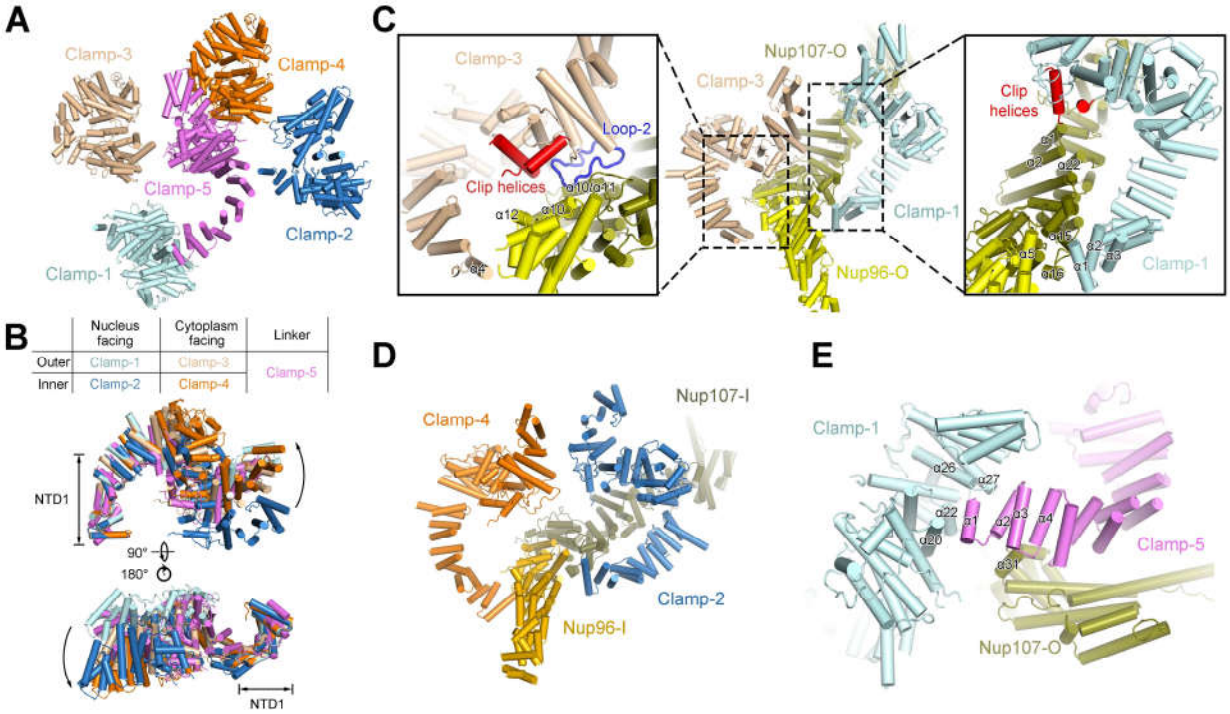


Fig. S13.

Structural analysis of Nup358 clamps. (**A**) An overview of the organization of the five Nup358 clamps in the CR subunit. (**B**) Conformational variations of the five Nup358 clamps. When aligned relative to NTD1, Nup358 clamps exhibit marked conformational shifts in their NTD2, as indicated by the black arrows. (**C**) Clamps-1 and -3 clench the outer Y complex. These interfaces are generally conserved between Clamps-2/-4 and inner Y complex as shown in Panel D. For Clamp-1, the Clip helices α24/25 are in close contact with the ridge of helices α1/2/22 in the Trunk of Nup107-O, and the N-terminal helices α1/2/3 interact with α15/16 from Nup107-O and α5 from Nup96-O. For Clamp-3, the Clip helices α24/25 directly contact helices α10/12 of Nup96-O and Loop-2 interacts with helices α10/α11 of Nup107-O. (**D**) Clamps-2 and -4 clench the inner Y complex. (**E**) Clamp-5 associates with Clamp-1 and grips the outer Y complex through its interface with Nup107-O. At the N-terminus of Clamp-5, helices α1 and α2 contact helices α20/22/26 and α27 from Clamp-1, respectively; helices α3/4 are in close proximity to helix α31 from Nup107-O.

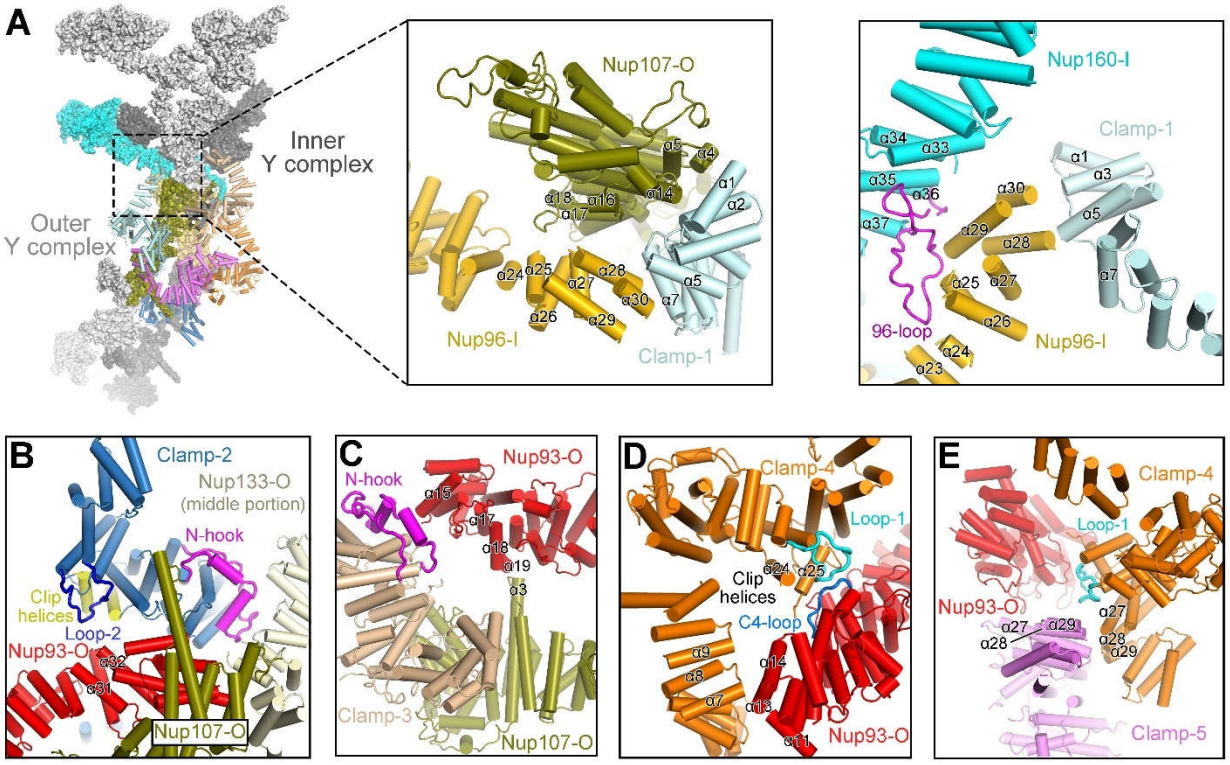


Fig. S14.

Nup358 clamps and Nup96-I mediate association between the two Y complexes within the same CR subunit. (**A**) Clamp1 and Nup107-O clench the Tail of Nup96-I. Helices α16/17/18 of Nup107-O associate with helices α24/25/27/28 of Nup96-I. Helices α3/α5 of Clamp-1 stack against helices α28/30 from Nup96-I. (**B**) Interfaces of Clamp-2 with Nup93-ACE1-O, Nup107-O and Nup133-O. Loop-2 and Clip helices of Clamp-2 associate with the ridge of helices α31/α32 from Nup93-O. The N-hook of Clamp-2 interacts with the ridges of helices from the middle portion of Nup133-O. (**C**) Interfaces between Clamp-3 and Nup93-ACE1-O. Helix α15 and the loop between α17 and α18 from the Crown of Nup93-ACE1-O interact with the N-hook of Clamp-3, and helix α19 of Nup93-ACE1-O contacts helix α3 of Nup107-O. (**D**) The interfaces between Clamp-4 and the Nup93-ACE1-O. At the Crown of Nup93-ACE1-O, helices α11/13/14 associate with helices α7/8/9 from the NTD1 of Clamp-4. At the Trunk of Nup93-ACE1-O, a flexible loop (named C4-loop, residues 233-241) contacts the Loop-1 and Clip helices of Clamp-4. (**E**) The interface of Clamp-5 with Nup93-ACE1-O and Clamp-4. At the C-terminus of Clamp-5, helices α27-α29 associate with α27-α29 from Clamp-4 and are in close proximity to the Trunk of Nup93-ACE1-O. These interfaces bridge clamps-1,3,5 that clench the outer Y complex with clamps-2,4 that clench the inner Y complex.

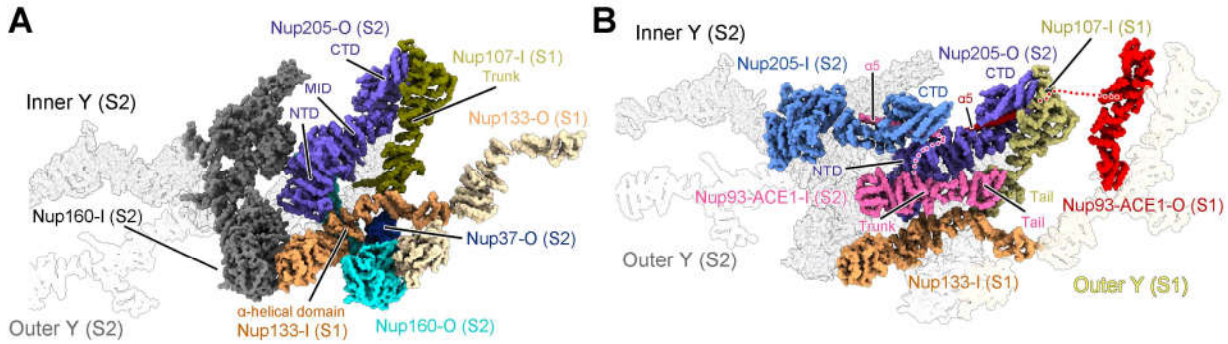


Fig. S15.

Nup205 and Nup93 play a central role in the inter-subunit association. (**A**) Nup205-O is a key bridge between neighboring CR subunits. Nucleoporins that map to the interface between the CR subunits are color coded and labeled. S1 and S2 refer to the neighboring CR subunits. (**B**) Nup93 molecules connect neighboring subunits. ACE1 of Nup93-I (S2) connects ACE1 of Nup107-I (S1) with the NTD of Nup205-O (S2). Nup93-α5 may sew the neighboring subunits by inserting into the axial groove of the Nup205-CTD. Similar views are shown in the two panels.

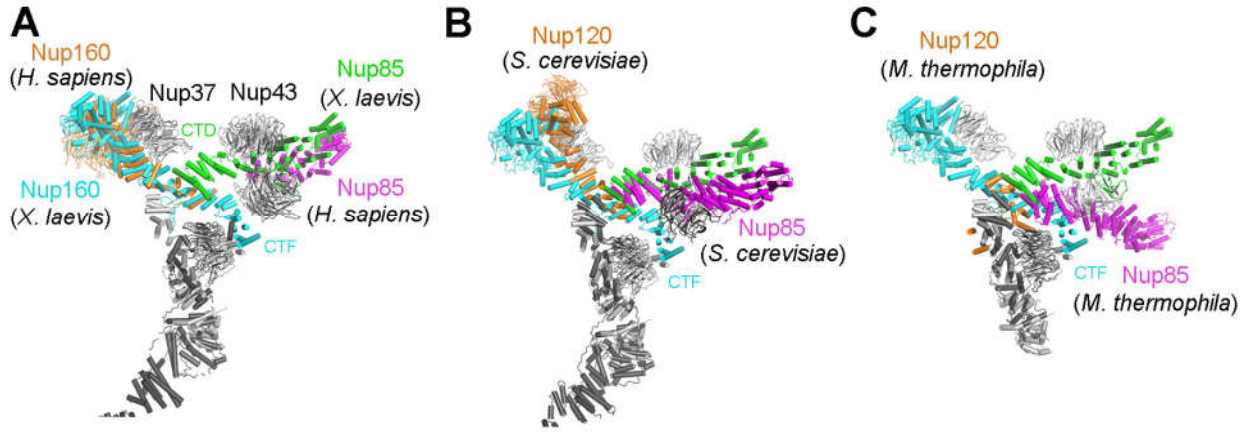


Fig. S16.

The C-terminal fragment (CTF) of Nup160 plays an important role in shaping the Y complex. (**A**) Structural comparison of the Y complexes from *X. laevis* and *H. sapiens*. Coordinates of the inner Y complex from this study are superimposed with the composite coordinates of the Y complex from *H. sapiens* (PDB code: 5A9Q) (*10*). The overall conformation and features of the Y complexes remain generally similar between these two structures, except for the following. First, Nup160-CTF (cyan) and Nup85-CTD (green) are resolved only in the *X. laevis* EM reconstruction. Second, the top and bottom faces of the Nup43 β-propeller in the *X. laevis* Y complex are opposite to those in the human Y complex. The two β-propellers, Nup37 and Nup43, both use their top face to bind to the α-helical domains of the short and long arms, respectively. (**B**) Structural comparison of the Y complexes from this study and from the *in situ* study of the *S. cerevisiae* NPC. Coordinates of the inner Y complex from *X. laevis* are superimposed with the composite coordinates of the Y complex from *S. cerevisiae* (*33*) (pdb file obtained from https://doi.org/10.5281/zenodo.3820319). (**C**) Structural comparison of the Y complexes from this study and the core from the *M. thermophila* Y complex (*19*). An apparent rotation of around 40 degrees of the short arm towards the stem-base is observed in the crystal structure of the *M. thermophila* Y complex core (PDB code: 4YCZ). Notably, both Seh1 and Nup160-CTF are absent in *M. thermophila*.

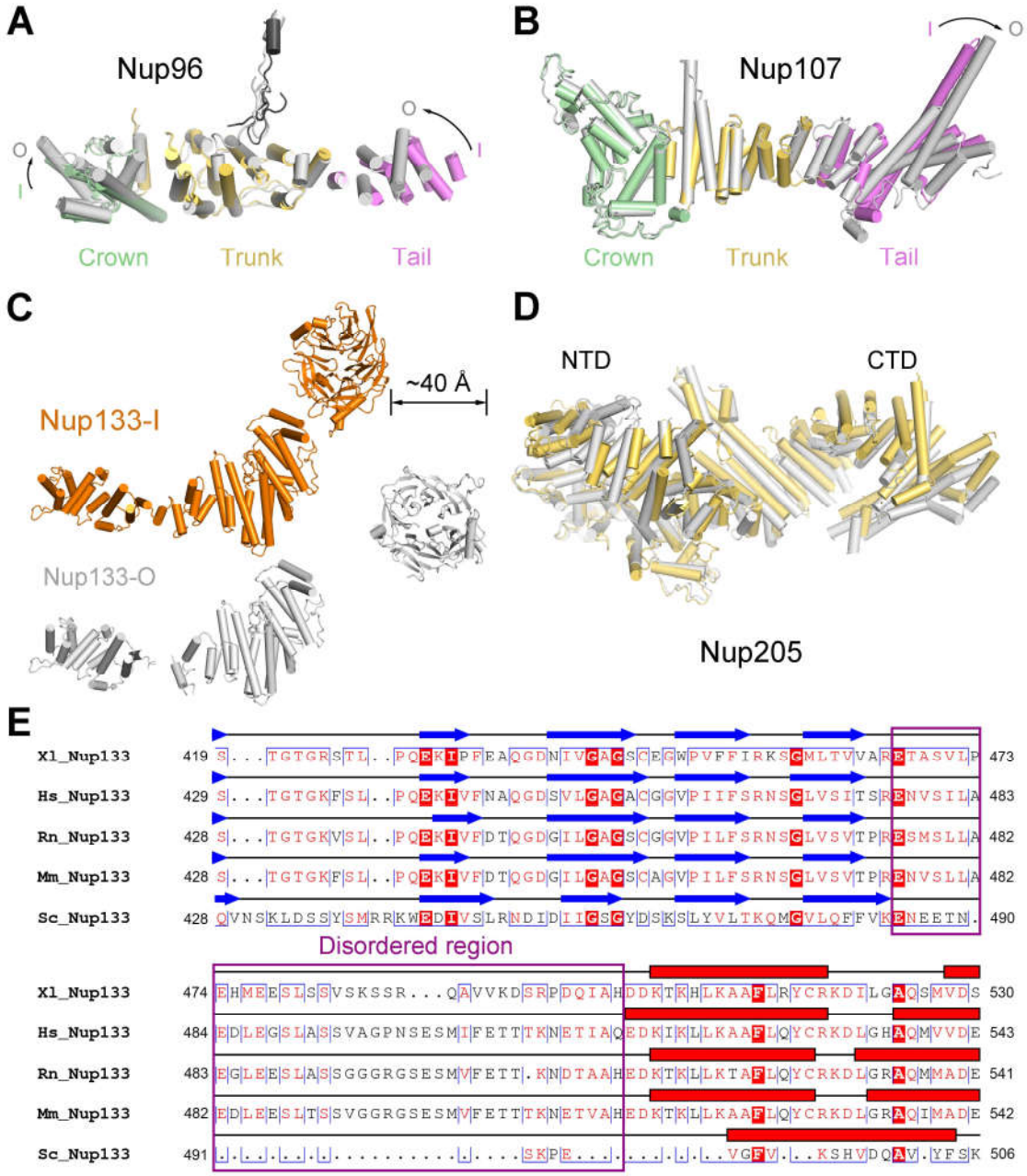


Fig. S17.

Structural elasticity of α-helical nucleoporins. (**A**) Structural comparison of Nup96 in the inner and outer Y complexes. The Crown, Trunk and Tail of Nup96-I are colored green, yellow and violet, respectively. Nup96-O is shown in gray. When the two Nup96 molecules are superimposed relative to their Trunk, which can be aligned with an RMSD of 1.8 Å over 209 Cα atoms, the other two domains, particularly the Tail domain, exhibit eminent domain-wise shifts. (**B**) Structural comparison of Nup107-I/O. The Crown, Trunk and Tail of Nup107-I complex are colored green, yellow and violet, respectively. Nup107-O is colored gray. When the Trunk domains of the two Nup107 molecules are superimposed, with an RMSD of 0.9 Å over 197 aligned Cα atoms, the Crown and Tail domains show conformational shifts with the Tail to a larger degree. (**C**) Structural comparison between Nup133-I/O. The N-terminal β-propeller of

Nup133-O is shifted by approximately 40 Å against its C-terminal α-helical domains when compared to that of Nup133-I. The two Nup133 molecules are aligned on their C-terminal α-helical domains, producing an RMSD of 2.1 Å over 526 aligned Cα atoms. (**D**) Structural alignment between Nup205-I (yellow) and Nup205-O (gray). When the two Nup205 molecules are aligned relative to their NTD (residues 13-922), the RMSD is 3.5 Å over 650 aligned Cα atoms. Meanwhile, the CTD exhibits marked shift. (**E**) Sequence alignment for the disordered region that links the N-terminal β-propeller and α-helical domain in Nup133. The predicted disordered region is absent in Nup133 from *Saccharomyces cerevisiae*. Xl, *Xenopus laevis*; Hs, *Homo sapiens*; Rn, *Rattus norvegicus*; Mm, *Mus musculus* and Sc, *Saccharomyces cerevisiae*.

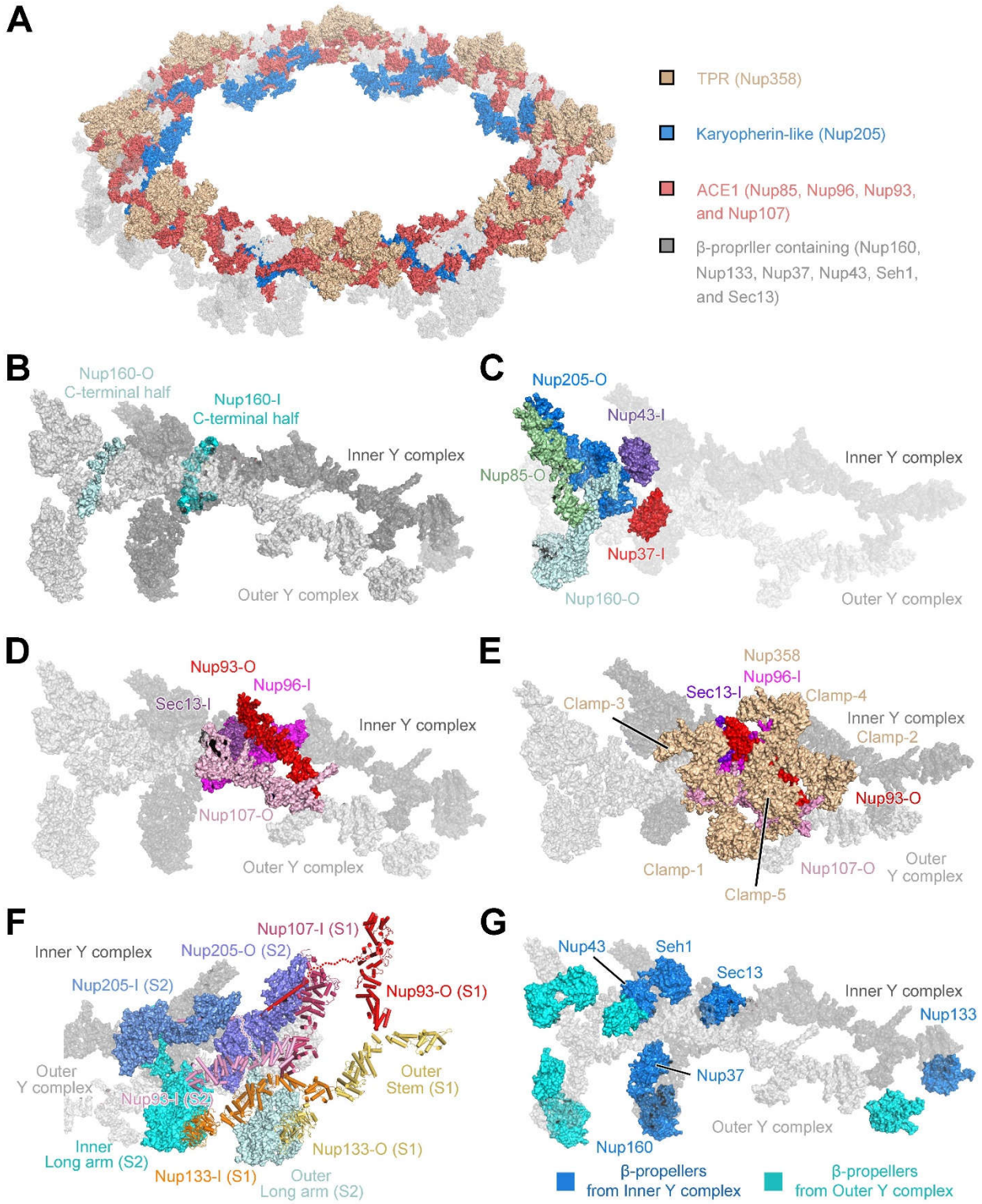


Fig. S18.

A summary of key components and interfaces in the assembly of the CR. (**A**) The CR scaffold is mainly constituted by the α-helical nucleoporins and fortified by the β-propellers. The ACE1 fold, Karyopherin-like fold and TPR fold α-helical nucleoporins are color coded. The membrane contacting proteins and β-propeller proteins are colored gray. (**B**) The C-terminal half

(α32 through α47) of Nup160 at the vertex connects the short arm and the stem base of Y complex by interacting with four nucleoporins (Nup85, Seh1, Nup96, Sec13). (**C**) Nup205-O connects both arms of the inner Y complex with the short arm and the vertex of the outer Y complex. Nup205-O directly binds Nup85-O, Nup160-O, Nup37-I, and Nup43-I. (**D**) Nup93-O, Nup107-O, Sec13-I, and Nup96-I form six pairwise interfaces in the stems of the Y complexes. (**E**) Nup358 clamps clench onto the stems of the Y complexes and bind Nup93-ACE1-O. (**F**) A close-up view on the inter-subunit interfaces. The stem of the inner Y complex (S1) associates with the long arms of both inner and outer Y complexes (S2). Within each of the two concentric rings, the β-propeller of Nup133 (S1) interacts with the long arm of the Y complex from the posterior subunit to form the head-to-tail architecture. The extended helix α5 of Nup93-O (S1) inserts into the axial groove of the CTD of Nup205-O (S2). ACE1 of Nup93-I (S2) connects Nup107-I (S1) and the NTD of Nup205-O (S2), while the helix α5 may reach into the axial groove of Nup205-I CTD (S2). (**G**) Twelve β-propellers solidify the structure of the two Y complexes.

Table S1. Cryo-EM data collection, refinement, and validation statistics.

	core region (7FIK) (EMD-36000)	Nup358 region (7FIK) (EMD-36001)	Nup358-NTF (7FIL) (EMD-36002)
Data collection and Processing			
Microscope	Titan Krios (Thermo Fisher Scientific)		
Voltage (keV)	300		
Camera	Gatan K3		
Magnification	64,000		
Pixel size at detector (Å/pixel)	1.387		
Total electron exposure ($e^-/Å^2$)	50/57/87/100		50
Exposure rate (e^-/pixel/sec)	19.5		
Number of frames collected during exposure	32/37/46/56		32
Defocus range (μm)	-1.0 ~ -4.0		
Automation software	AutoEMation2		
Tilt angle (°)	0/30/45/55		0
Energy filter slit width	20 eV		
Micrographs collected (no.)	10,040/7,557/14,107/14,439		17,147
Micrographs used (no.)	8,145/6,171/10,319/9,112		17,147
Total extracted particles (no.)	5,148,474		2,418,968
For each reconstruction:			
Refined particles (no.)	1,279,270		452,078
Final particles (no.)	1,279,270		452,078
Symmetry parameters	C1		
Estimated error of translations (pixels)/rotations (°)	1.06/1.42	1.39/2.55	1.64/3.16
Resolution (global, Å)			
FSC 0.5 (unmasked/masked)	8.5/4.0	9.6/5.7	4.1/3.6
FSC 0.143 (unmasked/masked)	4.6/3.7	7.5/4.6	3.3/2.9
Resolution range (local, Å)	50-3.3	50-4.0	50-2.7
Resolution range due to anisotropy (Å)	4.5-3.6	7.2-4.4	3.1-3.0
Map sharpening *B* factor ($Å^2$) / (B factor Range)	-120	-125	-100
Map sharpening methods	Ad-hoc B-factor		
Model composition			
Protein	yes		yes
Ligands	/		/
RNA/DNA	/		/
Model Refinement (for each model)			
Refinement package	Phenix		
- real or reciprocal space	real		
- resolution cutoff	3.7	4.7	3.0
Model-Map scores			
-CC	0.76	0.55	0.75
- Average FSC	3.7	4.7	4.0
B factors ($Å^2$)	174.49		99.94
Protein residues	19,037		537
Ligands	/		/
RNA/DNA	/		/
R.m.s. deviations from ideal values			
Bond lengths (Å)	0.008		0.004
Bond angles (°)	0.288		0.687
Validation			
MolProbity score	2.55		2.77
CaBLAM outliers	3.2%		3.1%

Clashscore	11.78	11.41
Poor rotamers (%)	4.29	7.80
C-beta deviations	0	0
Ramachandran plot		
Favored (%)	92.01	90.93
Outliers (%)	0.32	0

Table S2. Summary of model building for the CR subunit of *X. laevis* NPC.

	Molecule **Vertebrates**/ Yeast	Copy No.		Length *Xenopus*	UniProt No. *Xenopus*	PDB code	Modeling	Model length	Resolution (Å)	Chain ID Outer/Inner
Y complex	**Nup85**/Nup85	2		653	Q68FJ0	3F3F, 4XMM	HM	~600	4.0~4.5	B/b
	Nup160/Nup120	2		1435	A0A1L8GIX3	4GQ2, 4XMM	HM	~1100	4.0~4.5	E/e
			160-CTF	-	-	-	DM/AF	~230		
	Nup96/Nup145C	2		923	A0A1L8HBE3	3IKO, 4XMM	HM	~510	4.0~4.5	G/g
	Nup107/Nup84	2		916	A2RV69	3IKO, 3I4R	HM	~770	4.5~6.0	I/i
	Nup133/Nup133	2		1140	A0A1L8H1I9	1XKS, 3I4R	HM	~1000	4.5~6.0	J/j
	Sec13/Sec13	2		320	Q7ZYJ8	3BG1	HM	~280	4.0~4.5	H/h
	Seh1/Seh1	2		360	Q4FZW5	3F3F	HM	~310	4.0~4.5	D/d
	Nup43/-	2		375	Q05AW3	4I79	HM	~370	4.0~4.5	C/c
	Nup37/Nup37	2		326	Q66IZ6	4GQ2	HM	~320	4.0~6.0	F/f
Nup358	**Nup358**/ -	5	Clamp-1/2/3/4/5	2905	A0A1L8HGL2	-	HM/RD	~740	4.5~8.0	K/L/M/N/O
Nup93	**Nup93**/Nic96 (ACE1)	2		820	Q7ZX96	-	HM/AF	~590	4.5~6.5	P/p
	Nup93/Nic96 (α5)	2		820	Q7ZX96	-	AF	~50	5.0~6.5	S/s
Nup205	**Nup205**/Nup192	2		2011	Q642R6	5HB4	HM/RD	~1770	4.0~5.5	A/a
Nup88	**Nup88**/Nup82	1		728	Q4KLQ6	-	AF/RD	~400	6.0~8.0	Y
Nup155	**Nup155**/Nup170, Nup157	1		1388	F6UHT0	-	AF/RD	~270	5.0~7.0	W
Nup98/X	**Nup98/X**	1		819	A0A1L8HBE3	-	AF/RD	~150	6.0~8.0	X

Under the column labeled "Molecule", proteins from vertebrates and yeasts are shown, respectively, with vertebrate components in bold. Under the column labeled "Modeling", HM stands for homology modelling; DM stands for *de novo* modelling; AF stands for manual adjustment based on the AlphaFold model; RD stands for rigid docking.

Table S3. Summary of secondary structural elements in the structurally resolved nucleoporins of the core and Nup358 regions.

Protein	Full length (aa)	Residues modeled		Number of α-helices		Number of β-strands	
		Inner	Outer	Inner	Outer	Inner	Outer
Nup160	1435	1259	1195	47	46	29	29
Nup37	326	291	287	0	0	28	28
Nup85	653	533	539	27	27	3	3
Nup43	375	300	299	0	0	28	28
Seh1	360	282	307	0	0	24	24
Nup96	923	493	499	30	30	3	3
Sec13	320	286	291	1	1	24	24
Nup107	916	726	765	37	37	0	0
Nup133	1140	1028	1021	33	33	28	28
Nup205	2011	1604	1569	78	77	2	0
Nup93	820	607	644	33	33	2	2

		Residues modeled*					Number of α-helices/β-strands				
Nup358	2905	C1	C2	C3	C4	C5	C1	C2	C3	C4	C5
		703	699	654	704	683	32/0	32/0	29/0	32/0	32/0
Nup155	1388			269					15/0		
Nup88	728			339					2/29		
Nup98/X	819			161					5/11		

*C1~C5: Clamp-1~Clamp-5.

Table S4. Structural comparison of the CR components.

Inner Y	Outer Y	Full length	Optimal alignment (by PyMol)	
			RMSD (Å)	No. of aligned Cα
Nup37 (291)	Nup37 (287)	326	0.829	260
Nup43 (300)	Nup43 (299)	375	0.916	279
Sec13 (286)	Sec13 (291)	320	0.729	271
Seh1 (282)	Seh1 (307)	360	0.682	244
Nup85 (533)	Nup85 (539)	653	3.815	379
Nup96 (493)	Nup96 (499)	923	2.431	431
Nup160-CTF (152)	Nup160-CTF (166)	230*	5.006	110
Nup205 (1604)	Nup205 (1569)	2011	3.656	1331
Nup107 (726)	Nup107 (765)	916	2.759	614
Nup133 β-propeller (398)	Nup133 β-propeller (398)	1140	0.013	357
Nup133 α-helical domain (630)	Nup133 α-helical domain (623)	1140	2.108	526
Clamp-1 (703)	Clamp-2 (699)		2.366	543
Clamp-1 (703)	Clamp-3 (654)		0.687	491
Clamp-1 (703)	Clamp-4 (704)		5.567	695
Clamp-1 (703)	Clamp-5 (683)		4.721	614
Clamp-2 (699)	Clamp-3 (654)		4.239	536
Clamp-2 (699)	Clamp-4 (704)		4.087	579
Clamp-2 (699)	Clamp-5 (683)		7.404	600
Clamp-3 (654)	Clamp-4 (704)		0.763	485
Clamp-3 (654)	Clamp-5 (683)		2.705	491
Clamp-4 (704)	Clamp-5 (683)		5.077	562

*Only the C-terminal fragments (consisting of 230 residues) from Nup160-I and Nup160-O were compared.

Table S5. Summary of protein-protein interfaces in the CR.

Within the same CR subunit				Between two CR subunits	
Within the same Y complex		**Between two Y complexes**		Subunit-1	Subunit-2
Subject-1	Subject-2	Subject-1	Subject-2		
Nup85	Seh1	Nup96-I	Nup107-O	Nup133-O (β-propeller)	Nup160-O-NTD
Nup85	Nup43	Sec13-I	Nup107-O	Nup133-I (α-helical domain)	Nup160-O-NTD
Nup85-CTD	Nup160-CTD	Nup133-I	Nup133-O	Nup133-I (α-helical domain)	Nup37-O
Nup160 (96-loop)	Nup96-CTD	**Intra-CR subunit (except inter-Y complex)**		Nup133-I	Nup205-O
Nup160	Nup37	Subject-1	Subject-2	Nup107-I Finger helix	Nup85-O
Nup160-CTF	Sec13	Nup205-O Tower helix	Nup85-O	Nup107-I Finger helix	Nup205-O Tower helix
Nup160-CTF	Nup96-NTD	Nup205-NTD	Seh1	Nup107-I	Nup43-O
Nup160-CTF	Seh1	Nup205-O-TAIL	Nup85-O	Nup107-I	Nup205-O CTD
Nup96	Sec13	Nup205-O-NTD	Nup160-O-CTF	Nup93-O	Nup205-O
Nup96	Nup107	Nup205-O-NTD	Nup43-I		
Nup107	Nup133	Nup205-O-NTD	Nup37-I		
		Clamp-1	Nup107-O		
		Clamp-1	Nup96-I		
		Clamp-1	Clamp-5		
		Clamp-3	Nup96-O		
		Clamp-3	Nup107-O		
		Clamp-4	Nup96-I		
		Clamp-4	Nup93-O		
		Clamp-4	Clamp-5		
		Clamp-5	Nup107-O		
		Clamp-5	Nup93-O		
		Nup93-O	Clamp-3		
		Nup93-O	Nup107-O		
		Nup93-O	Nup96-I		
		Nup93-O	Sec13-I		
		Nup93-I	Nup205-I		